

TOKEY SIFULLAH TANJIL

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Google Scholar: scholar.google.com/citations?user=dAgMZWwAAAAJ

ResearchGate: <https://www.researchgate.net/profile/Tokey-Sifullah>



CAREER OBJECTIVE

To build a long-term career in academia, with a focus on research in applied mathematics.

Research Interest: Nonlinear Waves and Solitons; Plasma Physics and Plasma Fluid Dynamics; Mathematical Modelling; Nonlinear Dynamical Systems; Fractional-Order Differential Equations; Neurodynamics and Computational Neuroscience; Data-Driven Modelling and Machine Learning.

EDUCATION

M.Sc. in Applied Mathematics

2022 (result published 2025)

University of Chittagong — CGPA 3.73/4.00, 1st Class, 2nd Position

B.Sc. (Honours) in Mathematics

2021 (result published 2023)

University of Chittagong — CGPA 3.48/4.00

Higher Secondary Certificate (HSC), Science

2017

Tejgaon College, Dhaka — GPA 4.25/5.00

Secondary School Certificate (SSC), Science

2015

Nakhalpara Hossain Ali High School, Dhaka — GPA 4.61/5.00

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant — Chittagong University of Science and Technology (CUET)

Dec 2025 – Present

Working under the supervision of Prof. Dr. M. G. Hafez on "A Fractal-Fractional Hybrid Neuron-Network Model for Neurodegenerative Disease Progression: Merging Hodgkin-Huxley Dynamics with Wilson-Cowan Mean-Field Theory." Also supports M.Sc. and Ph.D. students with their research.

Research Assistant — University of Chittagong

Nov 2025 – Mar 2026

Worked under the supervision of Assistant Prof. Shahrina Akter on "The Head-on Collision in Dust-Acoustic Solitary Waves in an Unmagnetized Dusty Plasma with Trapped Ion Distribution."

Teaching Assistant (Intern)

Sep 2023 – Jan 2024

Katirhat Mohila Degree College, Hathazari, Chittagong. Taught HSC first-year students and delivered Business Mathematics classes for the Management Department during a five-month internship.

Associate Editor, Competitive Math Booster

Nov 2022

Associate editor of a mathematics reference book for the BCS competitive examination (Copyright Registration No. 27404).

PUBLICATIONS

- Islam, M.A., **Tanjil, T.S.** & Hafez, M.G. (2026). *Cylindrical and spherical propagation of ion-acoustic overtaking multi-wave structures in the upper ionosphere of Venus*. Advances in Space Research. DOI: [10.1016/j.asr.2026.09.001](https://doi.org/10.1016/j.asr.2026.09.001)
- Akter, S., **Tanjil, T.S.** & Hafez, M.G. (2026). *Interacting dust-acoustic multi-soliton and periodic waves in strongly coupled dusty plasma with ion-drag force*. Scientific Reports, 16, 28115. DOI: [10.1038/s41598-026-57659-z](https://doi.org/10.1038/s41598-026-57659-z)
- Akter, S., **Tanjil, T.S.** & Hafez, M.G. (2026). *Propagation and interaction of ion-acoustic single, N-solitons, and periodic waves in Venus' upper ionosphere*. Advances in Space Research, 77(4), 5236–5253. DOI: [10.1016/j.asr.2025.12.043](https://doi.org/10.1016/j.asr.2025.12.043)

WORKING PAPERS & RESEARCH IN PROGRESS

1. **A Fractal-Fractional Hybrid Neuron-Network Model for Neurodegenerative Disease Progression: Merging Hodgkin-Huxley Dynamics with Wilson-Cowan Mean-Field Theory.** In progress, as part of current research assistantship at CUET under Prof. Dr. M. G. Hafez.
2. **Mathematical Model for Dengue Transmission: A Comparative Study for Urban and Rural Areas in Bangladesh.** Submitted to the proceedings of the 4th International Conference on Applied Mathematics in Science and Engineering (AMSE-2025), India; presented at the conference. With Prof. M. G. Hafez, CUET.
3. **The Head-on Collision in Dust-Acoustic Solitary Waves in an Unmagnetized Dusty Plasma with Trapped Ion Distribution.** In preparation, based on research assistantship at the University of Chittagong under Assistant Prof. Shahrina Akter.

CONFERENCE PRESENTATIONS

1. **4th International Conference on Applied Mathematics in Science and Engineering (AMSE-2025), India**

Presenter, 29 Aug 2025 — "Mathematical Model for Dengue Transmission: A Comparative Study for the Perspective of Urban and Rural Areas in Bangladesh"

2. **24th International Mathematics Conference (24IMC 2025), Bangladesh**

Presenter, 18 Dec 2025 — "Interacting Dust-Acoustic Multi-Soliton and Periodic Waves in Strongly Coupled Dusty Plasma with Ion-Drag Force"

TECHNICAL SKILLS

- **Mathematical Software:** Maple, MATLAB, Mathematica
- **Programming:** Python, Fortran, C++
- **Typesetting:** LaTeX
- **Design & Imaging:** Adobe Photoshop, Adobe Illustrator
- **Productivity:** MS Word, Excel, PowerPoint

ADDITIONAL TRAINING

Introduction to Computer Application Packages and Unicode Bangla

Aug – Oct 2022

Bangladesh Computer Council

AWARDS & ACTIVITIES

- Champion, CU Mathematics Department Soccer Team, 2021
- Champion, CU Mathematics Department Cricket Team, 2019

LANGUAGES

- Bangla — Native
- English — Fluent working proficiency

REFERENCES

Dr. Md. Golam Hafez

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and Technology
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Dr. Mohammed Aman Ullah

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Shahrina Akter

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Department of Mathematics
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